WORKED SOLUTION

STEP III 2011, Question 2

Pure • Rational Roots of Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q2

The key idea

Multiplying $\mathrm{f}(p/q) = 0$ by q^{n-1} clears every denominator except one: the leading term becomes p^n/q while everything else is an integer. So p^n/q must be an integer, and since p and q share no factor, that forces $q = 1$. After that, each part is a question about *integer* roots, and these can be ruled out by size, by divisibility or, neatly, by parity.

Solution

The opening result

Suppose $\mathrm{f}(p/q) = 0$ with p, q coprime integers and $q > 0$. Then

$$0 = q^{n-1}\,\mathrm{f}\left(\frac{p}{q}\right) = \frac{p^n}{q} + a_{n-1}p^{n-1} + a_{n-2}p^{n-2}q + \cdots + a_1 p\,q^{n-2} + a_0 q^{n-1},$$

because the term $a_k(p/q)^k$ becomes $a_k p^k q^{n-1-k}$, and $n - 1 - k \geq 0$ for every $k \leq n - 1$. Every term after the first is an integer, so

$$\frac{p^n}{q} = -\left(a_{n-1}p^{n-1} + \cdots + a_0 q^{n-1}\right)$$

is an integer; that is, q divides p^n.

Ask yourself Why does $q \mid p^n$ force $q = 1$?

If $q > 1$, it has some prime factor r. Then r divides p^n, and since r is prime it must divide p itself. But then r is a common factor of p and q greater than 1, which is impossible. So q has no prime factor, and as $q > 0$ we get

$$q = 1.$$

Hence every rational root p/q of $\mathrm{f}(x) = 0$ equals p, an integer. (Notice that this is where we use that f is *monic*: a leading coefficient a_n would leave $a_n p^n/q$, and we could only conclude $q \mid a_n$.)

Two quick facts will help below. If m is an integer root, then

$$m\left(m^{n-1}+a_{n-1}m^{n-2}+\cdots+a_1\right)=-a_0,$$

so m divides a_0; and $m\neq 0$ since $a_0\neq 0$.

Part (i)

The number $2^{1/n}$ is a root of $x^n-2=0$, which is monic with integer coefficients and $a_0=-2\neq 0$. If $2^{1/n}$ were rational, it would therefore be an integer m with $m^n=2$. But

$$1^n=1<2<4\leq 2^n,$$

so the positive real number $2^{1/n}$ lies strictly between the consecutive integers 1 and 2, and it cannot be an integer. Hence $2^{1/n}$ is irrational for every $n\geq 2$.

Part (ii)

Here $n=3$ and $a_0=1$. Any rational root is an integer m dividing 1, so $m=\pm 1$. But

$$1^3-1+1=1\neq 0,\qquad (-1)^3-(-1)+1=1\neq 0.$$

So $x^3-x+1=0$ has no rational roots.

(Alternatively: $m^3-m=(m-1)m(m+1)$ is a product including two consecutive integers, so it is even, and m^3-m+1 is odd, never zero. This parity idea is exactly what part (iii) needs.)

Part (iii)

The polynomial x^n-5x+7 is monic with integer coefficients and constant term $7\neq 0$, so any rational root is an integer m. We show that m^n-5m+7 is always odd.

- If m is even, then m^n and $5m$ are even, so m^n-5m+7 is even $-$ even $+$ odd, which is odd.
- If m is odd, then m^n and $5m$ are both odd, so m^n-5m is even and m^n-5m+7 is odd.

Either way m^n-5m+7 is odd, so in particular it is not 0. Hence $x^n-5x+7=0$ has no integer roots, and therefore no rational roots, for any $n\geq 2$.

(Checking the divisors of 7 instead would mean showing $m^n-5m+7\neq 0$ for $m=\pm 1,\pm 7$ and *every* n, which works but is messier; parity handles all n at once.)

Checks

- Try a polynomial that *does* have a rational root, such as $x^2-x-6=(x-3)(x+2)$: the roots 3 and -2 are integers, as promised, and they divide $a_0=-6$.

- Non-monic polynomials can have non-integer rational roots, e.g. $2x - 1$; this is why the leading coefficient 1 matters.
- Numerically, $2^{1/2} \approx 1.414$ and $2^{1/3} \approx 1.260$ sit between 1 and 2; and $x^3 - x + 1$ takes the value 1 at $x = \pm 1$. A computer search over 3000 random monic integer polynomials found only integer rational roots, and confirmed $m^n - 5m + 7$ is odd for $|m| \leq 200$, $2 \leq n \leq 14$.

Common slips

- Multiplying by q^n instead of q^{n-1}: you get $p^n = -q(\cdots)$, which also works, but you must then say that $q \mid p^n$; with q^{n-1} the fraction p^n/q makes this explicit.
- Asserting "$q \mid p^n$ so $q \mid p$" without saying why; the argument goes through a prime factor of q.
- In (iii), checking only a few values of n. The argument must work for all $n \geq 2$ simultaneously.

Beyond STEP

This is the *rational root theorem* in its monic form: a rational number that is a root of a monic integer polynomial is an integer. In the language of algebraic number theory, the integers are *integrally closed*, and numbers that are roots of monic integer polynomials are called algebraic integers. The parity trick in (iii) is the simplest case of reducing a polynomial modulo a prime, a powerful tool for proving that polynomials have no roots or are irreducible. Davenport's *The Higher Arithmetic* is a lovely next step, and Allenby's *Rings, Fields and Groups* develops the algebraic side.

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